Spatial Accessibility to Global Fund Clinics in Nigeria

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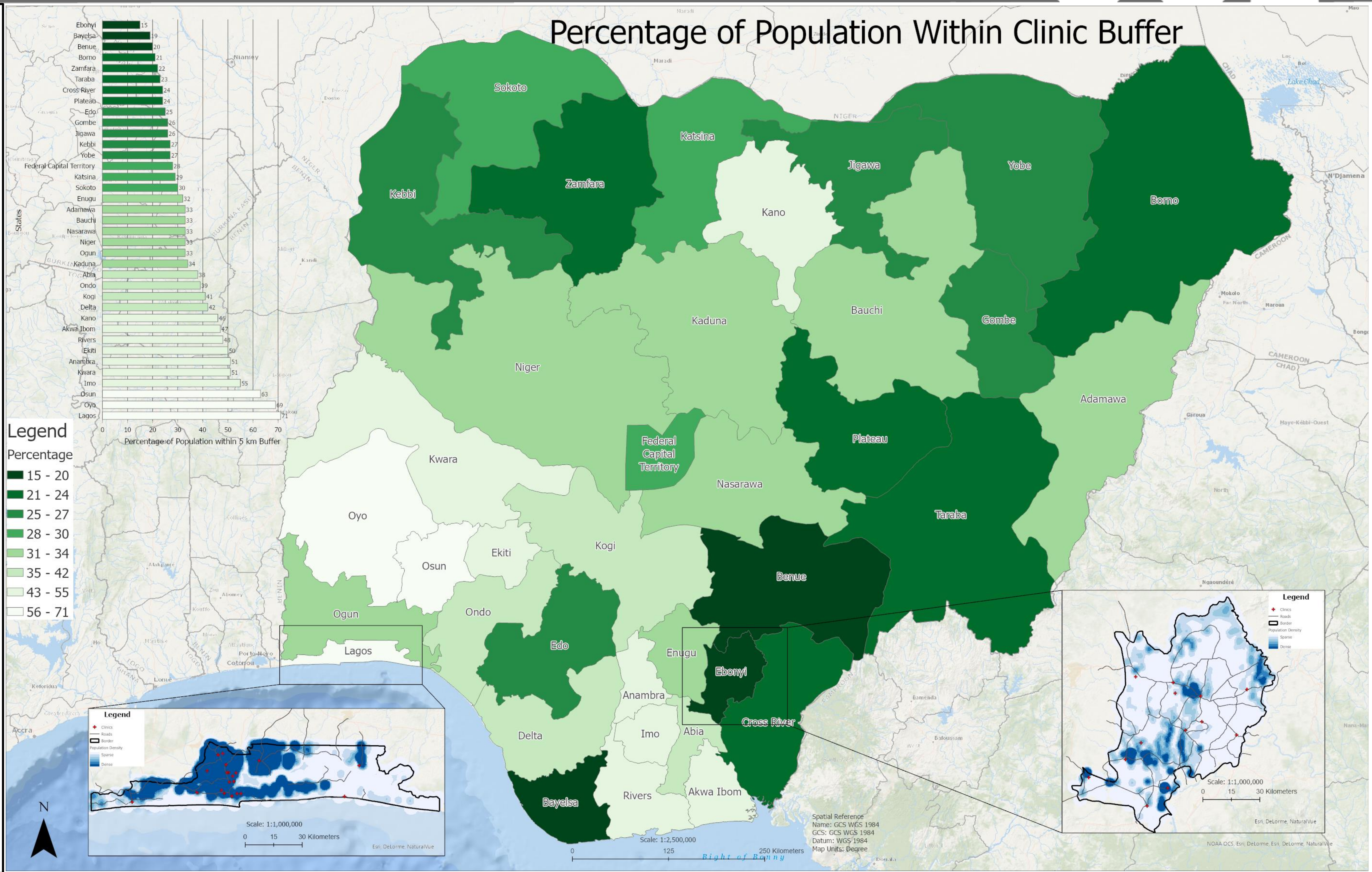
Introduction

Nigeria has some of the world's highest levels of HIV, Tuberculosis, and malaria. It has the greatest burden of malaria globally and has the highest levels of TB in Africa.[1] To combat this Nigeria has teamed up with The Global Fund an independent, multilateral, financing entity.

One way they are trying to tackle these problems is by distributing medication to local clinics covering all the states in Nigeria. This is a massive project that requires thoughtful positioning of clinics by where the high population density is, where high cases of each disease are, and other supply chain factors.

This map focuses on where the clinics are based, compared to population density, with the aim of showing how well each state's population is served by the clinics. Creating an easy to interpret map of this data will not only create awareness of the difficulties of distributing medicines to the whole of Nigeria, but also show trends in regions which are either more or less supported by Global Fund Clinics.



Evaluation of source data

Clinic Coordinate Data

The main data source for this project is the coordinates of the Global Fund Clinics. This information was in a spreadsheet which contained destination coordinates for the clinics which were used as the x and y coordinates for each clinic location. Due to the data being used to find the fastest route from central warehouses around Nigeria to clinics, various routes were suggested so there was a large amount of duplicate clinic points. To account for this, the duplicated points were removed using ArcGIS "Delete identical" feature. In addition, the x and y coordinate data had no reference to a coordinate reference system. Therefore WGS 1984 was chosen due to it being the predominant system used for our other datasets, and the majority of points appear to be in the correct place. However as seen in the clinic along the coast to the east of Lagos it is hard to tell if this is an accurate placement of the clinic or whether using the WGS 1984 coordinate system has distorted its placement.

Population Density

The Nigerian population density was downloaded from the Humanitarian Data Exchange branch of the United Nations Office for the Coordination of Humanitarian Affairs. The data is collected using machine learning to identify buildings from satellite data and combined with publicly available census data. [2]

Data reliability

Using machine learning to identify buildings is stipulated to give a 99.6% accuracy of positive labelling when comparing a building to a field or roads.[3] However, what is not mentioned in the study is how effective the machine learning algorithm is at detecting traditional African mud huts, also called a 'rondavel'. In Nigeria 14% of the total households are rondavels.[4] If the algorithm is not able to account for these structures, this will have a very large impact on the quality of our results.

Another reliability issue is the census data used for the density map. The last census taken in Nigeria was in 2006. Every year Nigeria adds 5 million to its population.[5] . Nigeria has a huge uncontrolled population growth due to high fertility rates caused by religious pressures for large families and lack of education on the benefits of smaller families. This means our population data will be about 75 million people short. In addition, the 2006 census has been criticised for not properly representing the southern states' population due to how funding is allocated by population per state. [6]

Methodology

To create the main map the Global Fund Clinic locations were converted from its csv to point data in WGS 1984 Coordinate system. In addition, the population density of the entire of Nigeria was also converted into point data in the same coordinate system and the boundaries for each state were obtained as polygons. Using the clinic locations, 5km buffers can be created from the clinics. The reason for a 5km buffer zone is only 16% of Nigerians have access to a motor vehicle so for these clinics to be accessible to the entire population they need to be close enough to easily walk to.[7]

The population density was then divided into a separate feature for each state to get accurate sum and points of the population for each state. Then using the population points with the buffers to give an accurate sum the points inside the clinic buffer zone. With these results the number of populations within 5km can then be divided by the total population of the state and times by 100 to gives us the population served by the clinic:

$$\frac{Number\ of\ People\ within\ Clinic\ Buffer}{Number\ of\ People\ in\ the\ Entire\ State} \times 100 = Percentage\ of\ People\ Served\ by\ Clinics\ in\ the\ State$$

One limitation of this approach is people who live on borders with other states might access clinics in the adjacent state that are within 5km from them.

Map Production/Cartography

For the map "Percentage of Population within Clinic Buffer" the percentage of population was used for each state. It was applied with gradated colours a pale green to dark green with light green being a high percentage and dark green being a low percentage covered. Jenks Distribution was used as it shows the trends in the data in the most effective way. Labels were added giving the name to each state as an easy way to refer to the chart and understand the spatial location. The colour scheme was taken from colour brewer to support people who are colour blind[8]. The Oceans base map was used to facilitate ease of location for the user.

For the bar chart the same colour scheme was used as the 'Percentage of Population within Clinic buffer' to allow quick understanding of the implications of a colour. In addition, the exact percentage is also labelled to allow the user to see the exact value and see the differences more precisely between each state.

The two regional maps were chosen due to Ebonyi being the worst served state and Lagos being the best served state. In these maps the point location of the clinics and the population density as a heat map are the main features. The population density is a graduated colour with a darker blue showing the higher density of points. The clinics are given a cross symbol and are red so can be quickly associated as places of care. The Oceans base map was used to help understand locations of clinics from the geography.

References:

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